How accurate LIGHT v0_3 is

The basic network of LIGHT v0_3, version 0.1.3, on 30 September 2026: v0_2 with the kinetic EEDF in the loop, with its electron side, its energy ladder and its checks

Bottom line. LIGHT v0_3 reproduces the fitted KrF decays [1, 2] within 2.3 to 6.1 % rms and predicts decays that no fit used [4] within 11.7 % rms, as v0_2 does, and at the Tahoma laser [5] it gives 88 % of the measured small-signal gain of 6.39 %/cm, where v0_2 gives 89 %.

Its electrons, solved kinetically during the run, land within 0.08 % of the gain and 0.08 % of the electron temperature of ML's self-consistent LIGHT-Boltzmann loop at Tahoma. It gives 111 % of Tahoma's saturation intensity and 99 % of its formation efficiency, and the C/B emission ratio of Yu and Setser [3] is matched to -0.5 %. No fit used Tahoma's numbers and no rate was set by a laser measurement.

Its largest misses are v0_2's, on data that conflict with the fitted sets, 19.3 and 34.1 % rms on the C-A decays of Hakuta et al. [6] and -22.3 % on the KrF(B) lifetime of Burnham and Searles [7], and the VUV yield of pure krypton at low pump, 132 % of Morozov's measurement [8].

[1] Eden, Waynant, Searles and Burnham, Appl. Phys. Lett. 32, 733 (1978). [2] Gadomski, Xu, Setser and Nelson, Chem. Phys. Lett. 189, 153 (1992). [3] Yu and Setser, J. Phys. Chem. 94, 2934 (1990). [4] Xu, Gadomski and Setser, J. Chem. Phys. 99, 2591 (1993). [5] Spectra Technology, *Experimental Support for KrF Laser Modeling*, final report to Los Alamos, 19 July 1989, OSTI 10136715. [6] Hakuta et al., Jpn. J. Appl. Phys. 29, 1194 (1990). [7] Burnham and Searles, J. Chem. Phys. 67, 5967 (1977). [8] Morozov, Heindl, Krücken, Ulrich and Wieser, J. Appl. Phys. 103, 103301 (2008).

LIGHT v0_3 is v0_2 with the kinetic EEDF in the loop. Its chemistry, its bands and its input are v0_2's, and the v0_2 overview describes them. What changes is where the electron rates come from. v0_2 integrates every electron rate over a Maxwellian at T_e with a tail read from a table. v0_3 solves the two-term Boltzmann equation with electron-electron collisions during the run, at LIGHT's own mean energy, and takes every electron rate, the elastic loss and the energy a captured electron carries from that solution. The 10 kinetic datasets have no electrons, so v0_3 scores them exactly as v0_2, and `make_overview_v0_3.py` refuses to write a number that differs. The tests with electrons are Tahoma's, the pure-krypton yield and the electron closure, and Table 1 lists every test.

1 What v0_3 changes

The kinetic EEDF comes from `v0_3/boltzmann.py`, ML's two-term scheme with its electron-electron term, written on sparse matrices. It carries exactly the electron processes the basic network books, so its energy balance and LIGHT's ledger close the same budget. Penning ionization injects its electrons at the energy the network gives them, Kr_2^+ recombination removes them with $\sigma v \propto \varepsilon^n$, the template of its T_e^n law, and the growth term carries the net. A beam's secondary electrons are injected with the sub-excitation spectrum of the degradation cascade, as ML's beam-sustained solver injects them. The elastic and field powers LIGHT reads are the parts of the solver's own Scharfetter-Gummel fluxes, so they are the energy it exchanges.

Every solution has LIGHT's mean energy, $\frac{3}{2}T_e$, held by a field $E/N \geq 0$, which in the pulse stands for the power the electrons take and after it for the energy they still carry. Where even the solution without a field is hotter than LIGHT's electrons, the field is 0. LIGHT reads a table of the kinetic rates, captured energies and elastic power at five mean energies around its own, renewed when any input of the kinetic state drifts, and past the table's ends it carries them with its own Maxwellian dependence on T_e. In the pulse the EEDF is advanced in time, one backward-Euler step from the last update's to LIGHT's mean energy, with the population's own change in the growth term. After the pulse it is solved steady, and there the growth term is dropped where it cannot give LIGHT's mean energy, where it moves more energy than the heating brings and the gas does not relax the shape first, and where it has no positive solution. `v0_3/README.md` gives the design and what was found on the way.

test	measured	v0_3	v0_2	role of the data
energy ledger closure		below 10^{-13} of E_{dep} in 4 runs	below 10^{-12}	bookkeeping
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	the same	train
KrF(B,C) slow decay, Gadomski 1992 Figs. 1 and 2		rms 6.1 and 4.2 %	the same	train
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	the same	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	the same	held out
decay constants Xu 1993 printed, 50 and 100 Torr Kr	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	the same	structure check
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	the same	held out, in conflict
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	the same	held out, in conflict
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-8.4 %	the same	held out
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.55, $\times1.32$	0.57, $\times1.36$	check
electron closure at Tahoma, ML's kinetic loop	T_e 1.610 eV	1.609 eV	1.60 eV	reference
Tahoma small-signal gain g_0	6.39 %/cm	5.64, $\times0.88$	5.69, $\times0.89$	device, score only
the same, Kannari's two Kr_2^* estimates		6.05, $\times0.95$	6.10, $\times0.95$	device, score only
the same, Kr_2^* + F at the top of its band		6.18, $\times0.97$	6.23, $\times0.97$	device, score only
the same, trimer on $T_g^{-1/2}$		2.82, $\times0.44$	2.84, $\times0.44$	device, score only
Tahoma saturation intensity I_{sat}	1.4 ± 0.1 MW/cm^2	1.56, $\times1.11$	1.55, $\times1.11$	device, score only
Tahoma formation efficiency $g_0 I_{\mathrm{sat}}/P_{\mathrm{in}}$	0.27	0.266, $\times0.99$	0.268, $\times0.99$	device, score only

Table 1: LIGHT v0_3 against the data, beside v0_2, every band at its default unless a row says otherwise. A percentage is the deviation from the data, signed for a single value and rms for a set. A $\times$ factor is the model over the measurement. "The same" is checked: the generator stops if v0_3's number differs from v0_2's.

2 How the numbers were made

Every number in this note is written by `docs/make_overview_v0_3.py` from v0_3 and from `ML/`, and a test fails when those files no longer match the model. ML runs v0_3 when `LIGHT_VERSION` names it. As for v0_2, Tahoma and pure krypton run under the prototype beam drive, a test fixture in `docs/` and not part of the model, whose ion pairs, excited atoms and secondary-electron energy enter LIGHT's equations. Its secondary electrons now enter the kinetic equation as well. ML's Tahoma configuration is v0_2's, the default network with every band at its default and n_e free from a seed of 1.0×10^{10} cm^{-3}, now with the kinetic EEDF in the loop. Each run is a process of its own, and a Tahoma run takes minutes where v0_2 takes seconds.

3 The electron side at Tahoma

ML's kinetic loop runs LIGHT on its Maxwellian closure, scales each electron row by the ratio of a beam-sustained Boltzmann solution's rate to the Maxwellian's at the same mean energy, and iterates until the

factors stop changing. v0_3 does the same inside the run, at every moment. Over Tahoma's 200 to 400 ns window the two agree:

	T_e (eV)	n_e (cm^{-3})	g_0 (%/cm)	formation
ML's loop, 10 passes	1.610	6.49×10^{14}	5.632	0.2988
v0_3	1.609	6.52×10^{14}	5.637	0.2990
v0_3 against the loop, %	−0.08	+0.47	+0.08	+0.07
v0_2 against the loop, %	0.7	2.5	1.0	0.5

Table 2: v0_3's electrons against ML's self-consistent loop at Tahoma. v0_2's deviations are absolute, v0_3's signed.

In the window LIGHT's mean energy is the kinetic one at every update, held by a field of at most 0.13 Td, since the beam's secondaries alone heat the electrons almost to it. The two energy balances agree to 6.0×10^{-4} of the beam's electron power. Table 3 shows what the kinetic EEDF does to each electron row, as its rate over LIGHT's Maxwellian rate at the same T_e.

row	reference point, 502 ns, T_e 3.00 eV	Tahoma, 302 ns, T_e 1.61 eV
G1	0.076	0.070
C1a	0.013	4.8×10^{-3}
C1b	3.4×10^{-3}	7.0×10^{-4}
C1c	1.5×10^{-4}	1.2×10^{-5}
C2	2.0×10^{-5}	2.3×10^{-7}
ATT	0.58	0.96
DIS_a3	1.2	1.0
DIS_A1	1.1	0.97
VIB1	0.65	0.97
VIB4	0.69	0.98
R11p	0.87	0.98

Table 3: The kinetic rate over the Maxwellian rate at the same T_e, for each electron row the kinetic EEDF carries: G1 and C1a to C1c excite Kr, C2 ionizes it, ATT attaches to F_2, DIS_a3 and DIS_A1 dissociate it, VIB1 and VIB4 are its first and last vibrational rows and R11p is Kr_2^+ recombination.

4 Tahoma and the bands

Under the beam drive v0_3 gives g_0 = 5.64 %/cm, the saturation intensity 1.56 MW/cm^2 and $g_0 I_{sat}/P_{in}$ = 0.266, against v0_2's 5.69, 1.55 and 0.268. Each band alone at its two ends moves the window's gain as Table 4 shows. Of the bands, the trimer's $(T_g/300)^n$ spans the most, as in v0_2.

band	g_0 (%/cm), low to high end	over the measured g_0
$Kr_2^* + F \rightarrow KrF(B) + Kr$	5.64 to 6.18	×0.88 to ×0.97
the trimer's $(T_g/300)^n$	5.64 to 2.82	×0.88 to ×0.44
$Kr^* + Kr^* \rightarrow Kr^+ + Kr + e^-$	5.66 to 5.64	×0.89 to ×0.88
$Kr_2F^* + F_2$	5.64 to 5.84	×0.88 to ×0.91
$Kr_2F^* + F \rightarrow 2Kr + 2F$	5.64 to 5.81	×0.88 to ×0.91
$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$, 300 K	5.64 to 5.26	×0.88 to ×0.82
$Kr_2^* + F_2$, both channels	5.64 to 5.69	×0.88 to ×0.89
$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	5.64 to 5.64	×0.88 to ×0.88

Table 4: Each band alone at its two ends at Tahoma under v0_3, the others at their defaults, over the 200 to 400 ns window. With every band at its default these runs give 5.64 %/cm. ML's formation run, the same network with ML's formation channels added, gives the 5.64 of Table 1.

5 The reference point

At the notes' reference point, 1 atm, 0.5 % F_2, 300 kW cm^{-3} for 1 μs and run to 2 μs, v0_3 and v0_2 give these shares of the deposited energy:

	v0_3	v0_2	v0_3 over v0_2, %
248 nm, KrF B-X	4.79	4.67	+2.7
272 nm, KrF B-A	0.153	0.149	+2.7
275 nm, KrF C-A	0.735	0.712	+3.2
420 nm, Kr_2F^*	4.96	4.87	+1.8
147 nm, Kr_2^*	19.7	20.0	−1.8
all bands	30.3	30.4	
to Kr^*, gross	99.4	98.3	
to Kr^+	26.0	26.8	
KrF(B) formed	24.6	23.8	
elastic heating	5.3	6.2	
T_g at the end (K)	650	649	

Table 5: The reference point, percent of E_{dep} unless a row says otherwise. v0_2's numbers are v0_3's run on v0_2's closure, which is v0_2 (a test checks it).

The kinetic electrons spend less on elastic heating and ionization and more on exciting Kr, and more KrF(B) forms. Over the pulse's second half the field brings 1.006 to 1.031 times the input power. The excess is the table's renewal. Between renewals LIGHT runs on a table made at the last renewal's electron density and gas. So at each renewal its electrons sit off the fresh table's balance by the excess, and they regain that balance well before the next renewal. Renewed every 5 ns, the excess is at most 1.015. With LIGHT's own $n_e\, d\langle\varepsilon\rangle/dt$ at the renewal, less the step's own, the two balances agree to 2.9×10^{-5} of the input power. The run takes 355 kinetic solutions, with E/N between 0.98 and 43 Td. Renewing the table at half the drift moves no band by more than 3.9×10^{-5} of itself, renewing it every 5 ns by 6.8×10^{-5}, and doubling the energy cells by 1.6×10^{-4}.

6 The first nanosecond

In the reference pulse's first 1 ns the electrons heat from the gas temperature, and most of the input power goes into raising their mean energy, which no steady solution holds. Solved steady, as v0_3 solved it until 1 October 2026, the EEDF dropped the growth term at 86 of its 88 updates there, its field carried 0.02 to 0.19 of the input power, and its energy balance missed LIGHT's by up to 0.22 of it. In the pulse v0_3 now advances the EEDF in time, one backward-Euler step from the last update's to LIGHT's mean energy, with the population's own change in the growth term and no rule to drop it. In the first nanosecond the field then carries 0.98 to 1.07 of the input power, 0 of the 87 updates drop the growth term, and after 0.1 ns the two energy balances agree to 1.5×10^{-3}. Before that, while the electrons are thermal and a cell is as wide as kT, they agree to 0.07.

The bands barely notice. The steady first nanosecond moves no band at the reference point by more than 1.9×10^{-4} of itself, and LIGHT's own Maxwellian closure in it moves none by more than 9.6×10^{-4}, since only 1.0×10^{-3} of the pulse's energy enters then. At Tahoma a steady start moves the window's gain by 3.6×10^{-4} and n_e by 9.2×10^{-4}, and 80 of its first nanosecond's steady solutions missed LIGHT's mean energy, against 0 of the time steps.

7 Pure krypton

Under the beam drive pure krypton at Morozov's low pump gives a VUV yield of 0.55 with v0_3, $\times 1.32$ the measured 0.42, against v0_2's 0.57. At 300 kW cm^{-3} it gives 0.28, against v0_2's 0.27. The kinetic electrons do not close v0_2's gap at low pump, which stays either physics LIGHT lacks or a systematic error of the measurement.

8 The kinetic EEDF's own checks

- With ML's electron-electron quadrature the solver is ML's dense one (`ee.solve_field`) to 2.2×10^{-9} in F and 4.2×10^{-8} in every rate at 3 states with the growth term. v0_3's own quadrature moves F by 1.1×10^{-5} at most there.
- The growth frequency is the solution's own net frequency, and the energy a field and the Penning electrons bring equals what the collisions, the captures and the growth term take, to 2.0×10^{-6} in the pulse and 5.6×10^{-5} with no field. Without the growth term the net loss of electrons comes back in the first cell, and with that injection the balance closes to 2.9×10^{-4}, a residual first order in the cell width.
- With no drive the solution is the grid's Maxwellian at the gas temperature to 1.4×10^{-14}, and it exchanges no energy with the gas to round-off. Taken by central differences, the elastic power gave it a loss to the gas, and late in an afterglow LIGHT's electrons cooled below the kinetic solution's mean. With ML's quadrature, F constant in each cell, the undriven solution condenses into the first cell, to a mean energy of 0.029 eV on 600 cells and 0.007 eV on 2400, where the gas gives 0.0388 eV.
- 0.12 ns into the reference pulse, at 0.4 Td, the problem without the growth term has two solutions, 0.282 eV from a hot start and 0.042 eV from a cold one, and at a fixed mean energy it has one. The field that mean energy needs falls from 0.444 Td at 0.06 eV to 0.388 Td at 0.2 eV and rises to 0.473 Td at 0.5 eV.
- The beam's secondaries reach the grid with the injected mean energy to 1.6×10^{-5} of it.
- A time step too long to matter is the steady growth solution to 1.7×10^{-12}, and one too short to matter, with no field, leaves F where it was to 1.2×10^{-10}. With e-e left out, a picosecond's step from the gas's Maxwellian under a Joule power of 10^9 eV/s per electron raises the mean energy by what the field and the sources bring less what the collisions, the captures and the population's change take, to 2.8×10^{-5}. With e-e at thermal energies, where a cell is as wide as kT, the flux scheme's energy exchange holds only to 1.0×10^{-1} on 600 cells and 7.6×10^{-3} on 2400.

9 The energy ladder

v0_3's network is v0_2's, and so is its ladder (Figure 1). KrF(B) lies 5.54 eV below the ion pair $Kr^+ + F^-$ and 5.06 eV above Kr + F, and the 248-nm photon carries 5.00 eV of that.

10 Where v0_3 is not yet reliable

- Everything in v0_2's list, since the chemistry is v0_2's: Kr_2^* + F, the trimer above 425 K, Kr^* + 2Kr above its measured range, hot gas, ion-ion recombination near atmospheric pressure, the two Penning rates, Kr_2^*'s well, and Hakuta's and Burnham and Searles' conflicts.
- The steady shape after the pulse. There the kinetic EEDF is a steady solution, while the electrons' shape is set by their history, and v0_3 takes the steady shape at their mean energy. In the pulse it now follows the history (Section 6).
- The first cell. Where the growth term is dropped, after the pulse only, the net loss of electrons comes back cold in the first cell, which cools a solution at low field. v0_3 holds LIGHT's mean energy

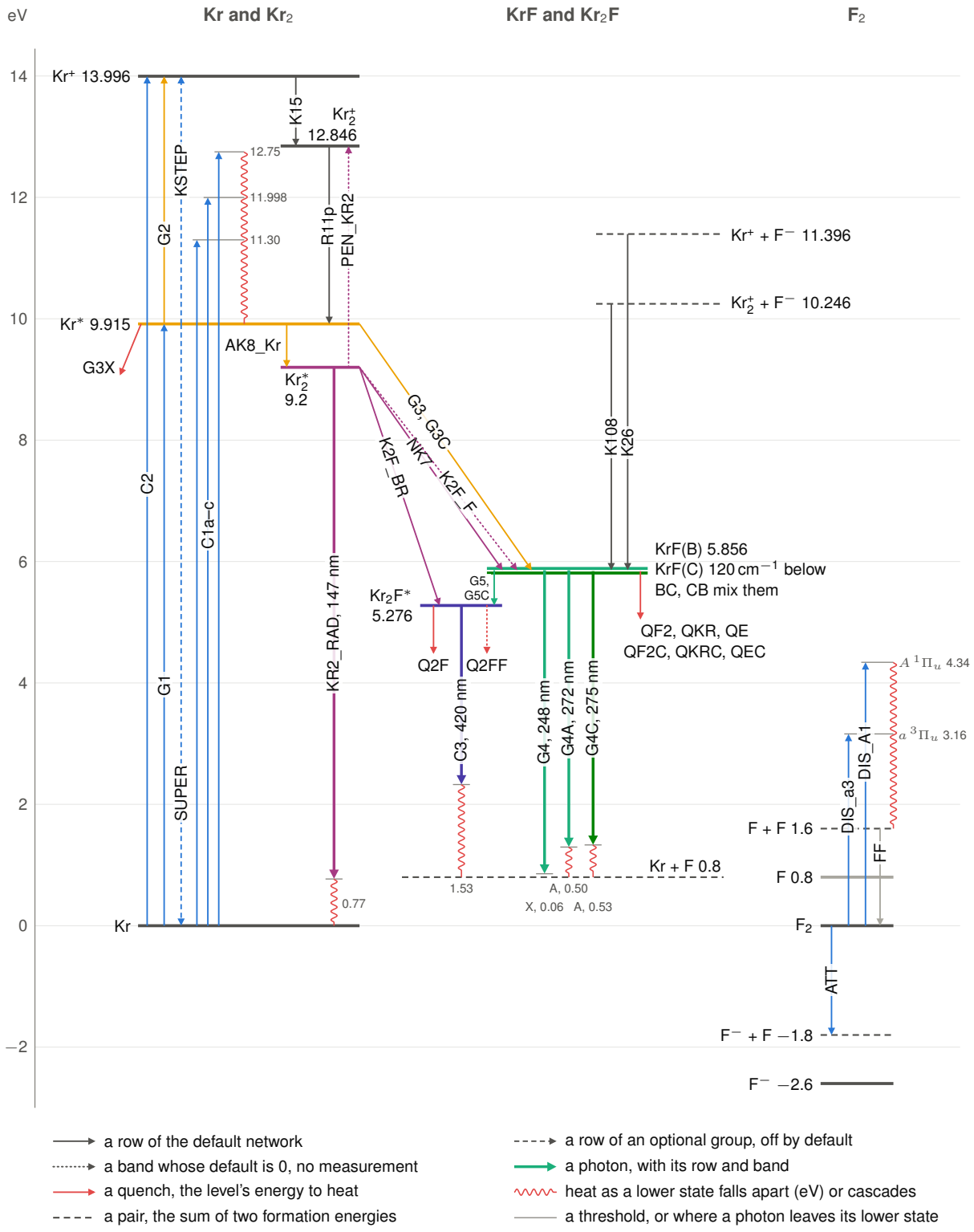


Figure 1: The energy ladder of LIGHT v0_3, version 0.1.3, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id. It is solid in the default network and dotted for a band whose default is 0, Q2FF, K2F_F and PEN_KR2. The rows of the optional group `electron` are dashed. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2 and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 120 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

regardless, so this moves the field and the low-energy shape.

- Thermal electrons. On the default 600 cells the grid's Maxwellian at 300 K has a mean energy of 0.0377 eV, below the 0.0388 eV of the gas, so late in an afterglow LIGHT's electrons sit a few percent below the gas temperature. 2400 cells give 0.0386 eV.
- Cost. A run takes minutes, so the bands' scans and the scan over the notes' ranges are hours under the kinetic closure.

A Reaction rates

v0_3's rows, rates and bands are v0_2's, and the v0_2 overview's Appendix A lists them. Its electron-impact rows run at the kinetic EEDF's rates, which Table 3 compares with the Maxwellian ones.